Product Description

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The fully automatic specific surface area and porosity analyzer adheres to ISO9277, ISO15901, international standards, and GB/T 19587 national standards. It tests the adsorption process based on the mass balance equation, static gas equilibrium, and pressure measurement using the static volumetric method. The testing process is conducted at liquid nitrogen temperature. When a known amount of gas is filled into the sample tube through a manifold, it causes a pressure drop, from which the molar mass of the adsorbed gas at adsorption equilibrium can be calculated. By measuring this equilibrium adsorption amount, theoretical models are used to determine the single-point and multipoint BET specific surface area, Langmuir specific surface area, BJH mesopore and macropore volume and area distribution, total pore volume, t-plot micropore volume and surface area, Dubinin-Astakhov micropore distribution, Horvath-Kawazoe and SF micropore distribution, MP micropore distribution, density functional theory (NLDFT), and Monte Carlo (GCMC) pore size distribution models, among other parameters.

Detailed Photos

Product Parameters

Specific surface area	0.0005m/g to no upper limit
Pore size analysis range	3.5 to 5000 angstroms(0.35-500 nanometers)
Test prnciple	Low-temperature nitrogen physisorption(static volumetric method)
Adsorption gas	Nitrogen(default).
Independent PO saturation pressure test tube,P/PO range 1×10-6-0.998	
Pressure and temperature measurementt	
Four imported absolute pressure sensors	ranging from 0 to 133KPa,with an accuracy of 0.1%.
Temperature sensor	PT-100,accuracy 0.1ºC,1 piece
Dewar flask	2L,made of metal,avoiding the fragility of glass Dewar flasks
Vacuum pump	Imported double-stage mechanical pump
Ultimate vacuum degree	1.0x10-4 Torr
Measurement software	adsorption/desorption isotherm determination

There are 2 analysis stations and 3 degassing stations,with a degassing temperature up to 400ºC.The analysis stations are immersed in a Dewar flask for simultaneous analysis,ensuring small dead volume,improving
analysis accuracy,and reducing operating costs.The three degassing positions are located in the same flexible
heating bag,and the analysis and degassing positions are independent.